

mdt data

To specify a range of addresses to be used in the data multicast distribution tree (MDT) pool, use the **mdt data** command in VRF configuration or VRF address family configuration mode. To disable this function, use the **no** form of this command.

mdt data threshold *kb/s*

no mdt data threshold *kb/s*

Syntax Description	threshold <i>kb/s</i> (Optional) Defines the bandwidth threshold value in kilobits per second (kb/s). The range is from 1 to 4294967.
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Command Default A data MDT pool is not configured.

Command Modes VRF address family configuration (config-vrf-af)
VRF configuration (config-vrf)

Command History	Release	Modification
	Cisco IOS XE Everest 16.5.1a	This command was introduced.

Usage Guidelines A data MDT can include a maximum of 256 multicast groups per MVPN. Multicast groups used to create the data MDT are dynamically chosen from a pool of configured IP addresses.

Use the **mdt data** command to specify a range of addresses to be used in the data MDT pool. The threshold is specified in kb/s. Using the optional **list** keyword and *access-list* argument, you can define the (S, G) MVPN entries to be used in a data MDT pool, which would further limit the creation of a data MDT pool to the particular (S, G) MVPN entries defined in the access list specified for the *access-list* argument.

You can access the **mdt data** command by using the **ip vrf** global configuration command. You can also access the **mdt data** command by using the **vrf definition** global configuration command followed by the **address-family ipv4** VRF configuration command.

Examples

The following example shows how to configure the range of group addresses for the MDT data pool. A threshold of 500 kb/s has been set, which means that if a multicast stream exceeds 1 kb/s, then a data MDT is created.

```
ip vrf vrf1
  rd 1000:1
  route-target export 10:27
  route-target import 10:27
  mdt default 236.1.1.1
  mdt data 228.0.0.0 0.0.0.127 threshold 500 list 101
!
.
.
!
ip pim ssm default
```

```
ip pim vrf vrf1 accept-rp auto-rp
!
```

Related Commands

Command	Description
mdt default	Configures a default MDT group for a VPN VRF.

snmp-server community

To configure the community access string to permit access to the Simple Network Management Protocol (SNMP), use the **snmp-server community** command in global configuration mode. To remove the specified community string, use the **no** form of this command.

snmp-server community [**clear** | **encrypted**] *community-string* [**view** *view-name*] [**RO** | **RW**] [**SDROwner** | **SystemOwner**] [*access-list-name*]
no snmp-server community *community-string*

Syntax Description	clear	(Optional) Specifies that the entered community-string is clear text and should be encrypted when displayed by the show running command.
	encrypted	(Optional) Specifies that the entered <i>community-string</i> is encrypted text and should be displayed as such by the show running command.
	<i>community-string</i>	Community string that acts like a password and permits access to the SNMP protocol. The maximum length of the <i>community-string</i> argument is 32 alphabetic characters. If the clear keyword was used, <i>community-string</i> is assumed to be clear text. If the encrypted keyword was used, <i>community-string</i> is assumed to be encrypted. If neither was used, <i>community-string</i> is assumed to be clear text.
	view <i>view-name</i>	(Optional) Specifies the name of a previously defined view. The view defines the objects available to the community.
	RO	(Optional) Specifies read-only access. Authorized management stations are able only to retrieve MIB objects.
	RW	(Optional) Specifies read-write access. Authorized management stations are able both to retrieve and to modify MIB objects.
	SDROwner	(Optional) Limits access to the owner service domain router (SDR).
	SystemOwner	(Optional) Provides system-wide access including access to all non-owner SDRs.
	<i>access-list-name</i>	(Optional) Name of an access list of IP addresses allowed to use the community string to gain access to the SNMP agent.
Command Default	By default, an SNMP community string permits read-only access to all MIB objects. By default, a community string is assigned to the SDR owner.	
Command Modes	Global configuration	
Command History	Release	Modification
	Cisco IOS XE Everest 16.5.1a	The command was introduced.
Usage Guidelines	To use this command, you must be in a user group associated with a task group that includes appropriate task IDs. If the user group assignment is preventing you from using a command, contact your AAA administrator for assistance.	

Use the **snmp-server community** command to configure the community access string to permit access to SNMP.

To remove the specified community string, use the **no** form of this command.

Use the **clear** keyword to specify that the clear text community string you enter is displayed encrypted in the **show running** command output. To enter an encrypted string, use the **encrypted** keyword. To enter a clear text community string that is not encrypted by the system, use neither of these keywords.

When the **snmp-server community** command is entered with the **SDROwner** keyword, SNMP access is granted only to the MIB object instances in the owner SDR. When the **snmp-server community** command is entered with the **SystemOwner** keyword, SNMP access is granted to all SDRs in the system.



Note In a non-owner SDR, a community name provides access only to the object instances that belong to that SDR, regardless of the access privilege assigned to the community name. Access to the owner SDR and system-wide access privileges are available only from the owner SDR.

Examples

This example shows how to assign the string comaccess to SNMP, allowing read-only access, and to specify that IP access list 4 can use the community string:

```
RP/0/RP0/CPU0:router(config)# snmp-server community comaccess ro 4
```

The following example shows how to assign the string mgr to SNMP, allowing read-write access to the objects in the restricted view:

```
RP/0/RP0/CPU0:router(config)# snmp-server community mgr view restricted rw
```

This example shows how to remove the community comaccess:

```
RP/0/RP0/CPU0:router(config)# no snmp-server community comaccess
```

Related Commands

Command	Description
snmp-server view	Creates or updates an SNMP view entry.

random-detect dscp

To change the minimum and maximum packet thresholds for the differentiated services code point (DSCP) value, use the **random-detect dscp** command in QoS policy-map class configuration mode. To return the minimum and maximum packet thresholds to the default for the DSCP value, use the **no** form of this command.

random-detect dscp *dscp-value* **percent** *min-threshold* *max-threshold*
no random-detect dscp *dscp-value* **percent***min-threshold* *max-threshold*

Syntax Description

<i>dscp-value</i>	The DSCP value. The DSCP value can be a number from 0 to 63, or it can be one of the following keywords: af11 , af12 , af13 , af21 , af22 , af23 , af31 , af32 , af33 , af41 , af42 , af43 , cs1 , cs2 , cs3 , cs4 , cs5 , cs7 , ef , or rsvp .
<i>percent</i>	Specifies that the minimum and threshold values are in percentage.
<i>min-threshold</i>	Minimum threshold in number of packets. The value range of this argument is from 1 to 512000000. When the average queue length reaches the minimum threshold, Weighted Random Early Detection (WRED) randomly drop some packets with the specified DSCP value.
<i>max-threshold</i>	Maximum threshold in number of packets. The value range of this argument is from the value of the <i>min-threshold</i> argument to 512000000. When the average queue length exceeds the maximum threshold, WRED or dWRED drop all packets with the specified DSCP value.

Command Modes

QoS policy-map class configuration (config-pmap-c)

Command History

Release	Modification
Cisco IOS XE Everest 16.5.1a	This command was introduced.

Usage Guidelines

Use the **random-detect dscp** command in conjunction with the **random-detect** command in QoS policy-map class configuration mode.

The **random-detect dscp** command is available only if you specified the *dscp-based* argument when using the **random-detect** command in interface configuration mode.

Specifying the DSCP Value

The **random-detect dscp** command allows you to specify the DSCP value per traffic class. The DSCP value can be a number from 0 to 63, or it can be one of the following keywords: **af11**, **af12**, **af13**, **af21**, **af22**, **af23**, **af31**, **af32**, **af33**, **af41**, **af42**, **af43**, **cs1**, **cs2**, **cs3**, **cs4**, **cs5**, **cs7**, **ef**, or **rsvp**.

On a particular traffic class, eight DSCP values can be configured per traffic class. Overall, 29 values can be configured on a traffic class: 8 precedence values, 12 Assured Forwarding (AF) code points, 1 Expedited Forwarding code point, and 8 user-defined DSCP values.

Assured Forwarding Code Points

The AF code points provide a means for a domain to offer four different levels (four different AF classes) of forwarding assurances for IP packets received from other (such as customer) domains. Each one of the four AF classes is allocated a certain amount of forwarding services (buffer space and bandwidth).

Within each AF class, IP packets are marked with one of three possible drop precedence values (binary 2{010}, 4{100}, or 6{110}), which exist as the three lowest bits in the DSCP header. In congested network environments, the drop precedence value of the packet determines the importance of the packet within the AF class. Packets with higher drop precedence values are discarded before packets with lower drop precedence values.

The upper three bits of the DSCP value determine the AF class; the lower three values determine the drop probability.

Examples

The following example enables WRED to use the DSCP value 8. The minimum threshold for the DSCP value 8 is 20, the maximum threshold is 40, and the mark probability is 1/10.

```
random-detect dscp percent 8 20 40
```

Related Commands

Command	Description
random-detect	Enables WRED
show queueing	Lists all or selected configured queueing strategies.

description (service template)

To add a description to a service template, use the **description** command in service template configuration mode. To remove the description, use the **no** form of this command.

description *description*
no description *description*

Syntax Description

<i>description</i>	Description of the service template.
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Command Default

A description does not display for the service template.

Command Modes

Service template configuration (config-service-template)

Command History

Release	Modification
Cisco IOS XE Release 3.2SE	This command was introduced.

Usage Guidelines

Use the **description** command to provide additional information about the service template when you display the service template configuration.

Examples

The following example shows how to configure a service template with a description:

```
service-template SVC_2
description label for SVC_2
access-group ACL_2
redirect url http://www.cisco.com
inactivity-timer 15
tag TAG_2
```

Related Commands

Command	Description
show service-template	Displays information about service templates.

match client-type

To create a condition that evaluates true based on an event's device type, use the **match client-type** command in control class-map filter configuration mode. To create a condition that evaluates true if an event's device type does not match the specified device type, use the **no-match client-type** command in control class-map filter configuration mode. To remove the condition, use the **no** form of this command.

match client-type {data | switch | video | voice}
no-match client-type {data | switch | video | voice}
no {match | no-match} **client-type** {data | switch | video | voice}

Syntax Description

data	Specifies a data device.
switch	Specifies a switch device.
video	Specifies a video device.
voice	Specifies a voice device.

Command Default

The control class does not contain a condition based on the device type.

Command Modes

Control class-map filter configuration (config-filter-control-classmap)

Command History

Release	Modification
Cisco IOS XE Release 3.2SE	This command was introduced.

Usage Guidelines

The **match client-type** command configures a match condition in a control class based on an event's device type. A control class can contain multiple conditions, each of which will evaluate as either true or false. The control class defines whether all, any, or none of the conditions must evaluate true to execute the actions of the control policy.

The **no-match** form of this command specifies a value that results in an unsuccessful match. All other values of the specified match criterion result in a successful match. For example, if you configure the **no-match client-type voice** command, all device values except voice are accepted as a successful match.

The **class** command associates a control class with a control policy.

Examples

The following example shows how to configure a control class that evaluates true if the client type is data:

```
class-map type control subscriber match-all CLASS_1
 match client-type data
```

Related Commands

Command	Description
class	Associates a control class with one or more actions in a control policy.
policy-map type control subscriber	Defines a control policy for subscriber sessions.

arp

To display the contents of the Address Resolution Protocol (ARP) table, use the **arp** command in boot loader mode.

arp [*ip_address*]

Syntax Description	<i>ip_address</i> (Optional) Shows the ARP table or the mapping for a specific IP address.
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Command Default	No default behavior or values.
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Command Modes	Boot loader
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Command History	Release	Modification
	Cisco IOS XE Everest 16.5.1a	This command was introduced.

Usage Guidelines	The ARP table contains the IP-address-to-MAC-address mappings.
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Examples	This example shows how to display the ARP table:
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```
Device: arp 172.20.136.8  
arp'ing 172.20.136.8...  
172.20.136.8 is at 00:1b:78:d1:25:ae, via port 0
```